A Simple Rule Beats a Solar Plant's Own Plan

Solar Yield Predictor: a day-ahead energy forecast for the Otrar LLP solar plant

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FRP 2026, AI Engineering, ISSAI, Nazarbayev University

October 2026

Abstract

A solar plant must send the grid a plan of how much energy it will make, and every missed kilowatt-hour is an imbalance. We used five months of real files from the Otrar LLP solar plant (Kargaly, Turkestan Region) to compare the plant's own plan with simple rules and with weather-based models. On 26 October days that no method had seen, the simple rule "tomorrow = today" was 83.2% accurate per day and the plant's plan 75.0%, so our daily miss was about one third smaller (1,110 vs 1,651 kWh), and our rule was closer than the plan in 5 of 5 months. A forecast does not need a complex model to help: the plant's own recent output already tells a lot about tomorrow.

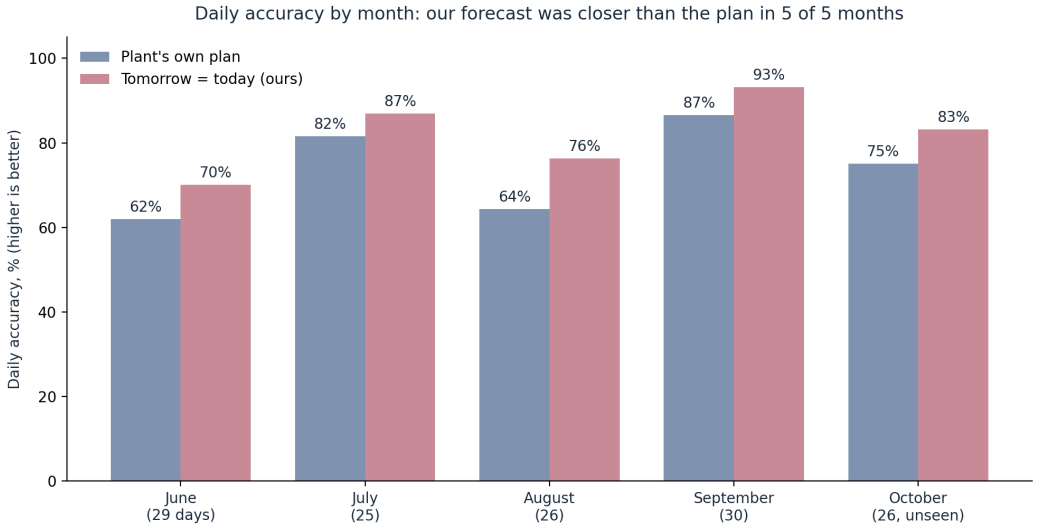


Figure 1: Daily accuracy by month (days with an outage or a zero plan are left out). Our forecast "tomorrow = today" was closer to the real energy than the plant's own plan in all five months. October was the unseen test.

1 Introduction

A solar plant sells its energy to the grid. Before each day, it must send a plan: how many kilowatt-hours it will make in every hour. When the real energy differs from the plan, the grid must cover the difference (an imbalance). A better plan means less imbalance.

The Otrar LLP plant gave us its real files: daily plan-fact tables, hourly daily reports and hourly plan cards. They show that the plan is often far from the real energy. On 2 September, for example, the plan was 5,192 kWh and the plant made 8,355 kWh. We asked a simple question: can a simple, honest forecast be closer to the real energy than the plan the plant really sent?

We cleaned 149 days of data, tested simple rules and weather models, and kept the test fair: every forecast is made only from data known before the day, and we chose our method before the final test month. Our contributions are:

- two clean tables from the plant's files: 149 days of plan and real energy, and 88 days hour by hour;
- a fair comparison of more than 10 forecasting methods with the plant's own plan, on days the methods never saw;
- a public web page that reads the plant's files in the browser and shows tomorrow's forecast in under one second.

2 Related work

Reviews of photovoltaic power forecasting [1] describe many methods, from physical models to machine learning, and use simple reference forecasts such as persistence ("tomorrow = today") to judge them. We use the same idea, but our main baseline is different: the real plan that the plant sent to the grid. The pvlib library [3] models the position of the sun; we used its clear-sky model to test the shape of a day. Our weather data comes from the Open-Meteo historical forecast API [4], so we only use weather forecasts that were known in advance. One of our models is a random forest [2].

3 What we built

Data. The plant's plan-fact tables give the daily plan and the real energy for June to September (122 days). Daily reports and plan cards give hourly real energy and the hourly plan for August, September and October (88 days, 2,112 hours); October has 27 days (1–27 October). The plant is in the village of Kargaly (42.866 N, 68.376 E). Two traps had to be solved: the plan cards are written in UTC+1, so we moved them 4 hours to Astana time (UTC+5), and a report day runs from 04:00 to 03:00 the next morning, with each hour written at its end. We checked our tables against the original Excel cells by hand. Days when the plant was off (outage) or the plan was almost zero are marked and counted apart.

Forecasts. We tested simple rules (yesterday's energy, the median of the last 7 days, the mean of the last 2–3 days), weather models from Open-Meteo forecasts (a straight line from sun and temperature, a correction of yesterday by the change in sun, a random forest), and a "cloud rule" that lowers yesterday's value when much less sun is forecast. The final daily forecast is the simplest rule: **tomorrow's energy = today's real energy** (if today was an outage, the last normal day). For hours, we split this daily total by the average shape of the last 7 normal days.

The page. A web page reads the plant's files in the browser (nothing is uploaded) and shows the forecast for the next day, by day and by hour, next to the plant's plan (Figure 2). It read 63 files in 0.2 seconds and 122 files in 0.64 seconds.

4 How we measure

For each day we compare a forecast with the real energy. The **daily miss** is the average of $|\text{forecast} - \text{real}|$ in kWh; the **miss in %** is the daily miss divided by the average real energy of the same days; **accuracy** $= 100\% -$ miss in %. We also count the days when the forecast was within 10% of the real value. For hours we use the same measures on the hours when the plant produces energy.

The **baseline** is the plant's own plan. Every forecast uses only data known before the day. Models learn only on earlier months. We chose our final rule after seeing July and September, so the clean test is **October**: the rule was fixed before we opened October, and we scored October once.

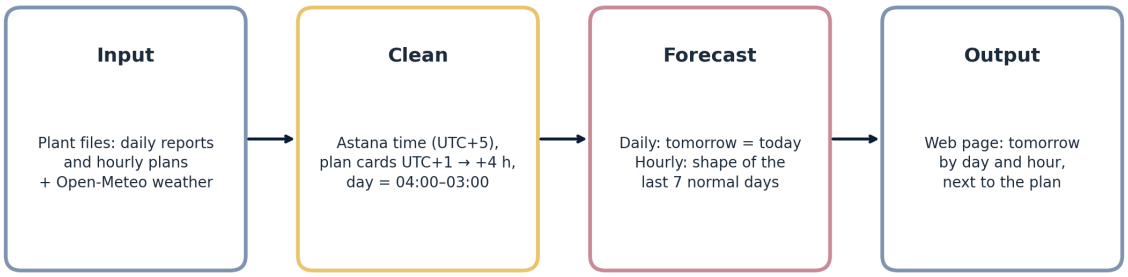


Figure 2: How it works: the plant's files and the weather → cleaning → forecast → a web page with tomorrow's forecast next to the plan.

5 Results

On the 26 normal October days (4 October was an outage), our rule was 83.2% accurate and the plant's plan 75.0% (Table 1). Our daily miss was 1,110 kWh against 1,651 kWh, about one third smaller, and our forecast was within 10% on 15 of 26 days against 10 of 26. Hour by hour, our accuracy was 80.9% against 72.3%. Two other simple rules (the 7-day median and the cloud rule) scored about the same as ours, 84.6–84.7%.

October, 26 normal days	Miss, kWh ↓	Accuracy ↑	Within 10% ↑	Hourly acc. ↑
Baseline: the plant's own plan	1,651	75.0%	10 of 26	72.3%
7-day median	1,012	84.7%	15 of 26	–
Cloud rule (30%)	1,019	84.6%	15 of 26	–
Ours: tomorrow = today	**1,110**	**83.2%**	**15 of 26**	**80.9%**

Table 1: The final test on 26 October days that no method had seen. One run; the rule was fixed before October. ↑ = higher is better, ↓ = lower is better.

In September (30 days), the same rule missed 513 kWh a day (93.2% accurate, within 10% on 23 of 30 days) and the plan 1,015 kWh (86.6%, 12 of 30). Even made 3 days in advance, the rule (88.3%) was more accurate than the plan. Our weather models did not win: in July, a straight-line weather model learned on June missed 975 kWh a day (79.6%), more than the plan (879 kWh); in September, the best weather models missed about 1,550 kWh (about 79–80%). They predicted too little in September, when the plant made more energy with less sun than in summer. Over all five months, our rule was closer than the plan every month (Table 2).

Month (normal days)	Plant plan: miss	Ours: miss	Ours: days within 10%
June (29)	38.0%	**29.9%**	9 of 29
July (25)	18.4%	**13.0%**	14 of 25
August (26)	35.7%	**23.7%**	9 of 26
September (30)	13.4%	**6.8%**	23 of 30
October (26)	25.0%	**16.8%**	15 of 26

Table 2: Daily miss in % of the average real energy, by month (lower is better). Only October is an unseen test.

6 Limitations

Our data is from one plant and five months (June to October); winter can be very different. We chose the rule after seeing September, so only October is a clean test, and it has 26 days. The rule cannot see a cloudy day

coming: on 30 September the plant made 4,631 kWh, and both our forecast (+59.6%) and the plan (+63.5%) were far too high. The plant sent some plans several days early (the plans for 2–4 September were sent on 31 August), while our main forecast is made one day early. We do not know the price of one kilowatt-hour of imbalance, so we report energy, not money.

7 Ethics

The data belongs to the Otrar LLP plant and was given to us for this project. It is not public: the Excel files and the day-by-day tables are not in our public code or on our pages; we publish only summary numbers and a few example days. The files contain no data about people that we use, and we removed all names. Weather data is from Open-Meteo. We used an AI coding agent to write code and to check the English; every number was checked against the files.

8 Conclusion

We cleaned five months of real data from a solar plant and compared more than 10 forecasts with the plant's own plan. On 26 unseen October days, the simple rule "tomorrow = today" was 83.2% accurate against 75.0% for the plan, and it was closer in all five months. Next, we want to test it on new months and to learn from the plant how early its plan must be sent and what changes its output.

Acknowledgments

We thank the Otrar LLP plant for its data, and our mentors at ISSAI and the FRP 2026 program.
Project page: https://zhureswe.github.io/solar-yield-predictor/
Live demo: https://kitan-a.com/frp/pub/s06/solar/
Code: https://github.com/zhureswe/solar-yield-predictor

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